

3: Scaling: min-max & Standard

(A) Min Max normalization:

[18, 42, 66, 90] min: 18 max: 90

$$X_{\text{norm}} = \frac{18-18}{90-18} = 0$$

$$X_{\text{norm}} = \frac{42-18}{90-18} = \frac{24}{72} = 0.333$$

$$X_{\text{norm}} = \frac{66-18}{72} = \frac{48}{72} = 0.666$$

$$X_{\text{norm}} = \frac{90-18}{90-18} = 1$$

Normalised [18, 42, 66, 90]



[0, 0.333, 0.666, 1]

(B) STD of [8, 10, 10, 14, 14, 16]

$$\bar{x} = \frac{8+10+10+14+14+16}{6} = \frac{72}{6} = 12$$

$$STD = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} =$$

Deviation

$$(8-12)^2 = -4^2 = 16$$
$$(10-12)^2 = -2^2 = 4$$
$$(10-12)^2 = -2^2 = 4$$
$$(14-12)^2 = 2^2 = 4$$
$$(14-12)^2 = 2^2 = 4$$
$$(16-12)^2 = 4^2 = 16$$
$$\sqrt{\frac{16+4+4+4+4+16}{6}} = \sqrt{\frac{48}{6}} = \sqrt{8} = 2.828 = STD$$

(C) Advantages and Disadvantages of min-max norm:

Advantage: Simple and easy to use ~~normalize~~ and normalize with.

Disadvantage: if the values are very far apart and most of the values are in the 10-20 range →